Bilateral gauge saturation on genuinely collapsed local covers

24 September 2026. This note connects the actual-form estimates to a collapsed regularity theorem. It does not replace base noncollapse by a renamed base-volume hypothesis. Instead, maximal local nilpotency rank supplies harmonic charts **after local rewinding**, and a covering-average estimate transfers the gauge information to those charts. The output is local product geometry outside a quantitatively small-volume set, not a global Einstein repair or a classification of all collapsed regions.

1. A collapse-tolerant structure statement

Use the bilateral actual conformal Spin^c inputs and conventions of bilateral realization, Section 1. Thus a_+ is actual in the given orientation and a_- in the opposite orientation; their positive projected energies satisfy $p_+, p_- > 0$. Define

$$\delta = \|\operatorname{Ric}^\sharp + I\|_{\infty,\mathrm{op}}, \quad r = \fint |\operatorname{Ric} + g|^2, \quad q_\pm = \frac{\frac{3}{2}\int (s^2 + 12|W^\pm|^2) - 72\pi^2 p_\pm}{V}, \quad K = 2r + q_+ + q_-. \tag{1}$$

The $q_\pm$ are nonnegative; Weyl norms are endomorphism norms. There is **no** lower bound on $p_\pm$, total volume, or the volume of base balls in the result below.

Fix a scale $0 < \rho \leqslant 1$. Let $\epsilon_N, h_N > 0$ be the four-dimensional Euclidean-model constants in Naber–Zhang's Proposition 6.4, reduced if needed so that $h_N < 1$. Define $\mathscr{G}_\rho$ to consist of points x for which, for some $k \in \{1,2,3,4\}$,

$$d_{GH}(B_{2\rho}(x), B_{2\rho}(0^k)) < \epsilon_N \rho, \qquad \operatorname{rank}\operatorname{Image}[\pi_1(B_{\epsilon_N\rho}(x)) \to \pi_1(B_{2\rho}(x))] = 4 - k. \tag{2}$$

The rank is the Hirsch rank of a finite-index nilpotent subgroup, not a Betti number or the rank of $\pi_1(M)$. The small group in (2) is almost nilpotent under these hypotheses. Closedness supplies the local completeness required by the classical theorem. A fixed measurable subset of the points satisfying (2) can be used if desired.

Let $\pi_x : U_x \to B_\rho(x)$ be the **local universal cover** with its pulled-back metric, and choose a lift $\tilde{x}$. The metric space U_x is generally incomplete at the lifted outer boundary. All balls used below stay a definite distance from that boundary. We do not identify U_x with a subset of the universal cover of M.

Theorem A (rewound product structure on the bulk). Fix $\rho \in (0,1]$ and $\alpha \in (0,1)$. There are $a > 0$, $C < \infty$, and a modulus $\Psi(t) \to 0$ as $t \downarrow 0$, depending only on ρ, α and the fixed classical constants, with the following property. Suppose M is a closed connected oriented smooth four-manifold with the actual inputs above, $\delta \leqslant 1/2$, and $0 < K \leqslant 1/2880$. Then there is a measurable set $\mathscr{E} \subset M$ satisfying

$$\frac{\operatorname{Vol}(\mathscr{E})}{V} \leqslant C K^{1/4}, \tag{3}$$

such that, for each $x \in \mathscr{G}_\rho \setminus \mathscr{E}$, a neighbourhood of $\tilde{x}$ containing its radius-a ball is $\Psi(\delta + K^{1/4})$-close in pointed geometric $C^{1,\alpha}$ to a neighbourhood in

$$\mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}. \tag{4}$$

One may define this closeness using a pointed embedding of a fixed slightly larger model ball, whose image contains $B_a(\tilde{x})$, and the scaled $C^{1,\alpha}$ norm of the difference of the pulled-back metrics. Shrinking a by a fixed factor makes these equivalent usual local definitions interchangeable.

Equivalently, along any sequence with $\delta_j \to 0$, $K_j \to 0$, there are exceptional sets of relative volume tending to zero, and all the indicated rewound balls converge uniformly in this local geometric sense to product balls. For $K = 0$, the equality case already makes g Einstein and Kähler in both orientations, so the product statement is exact.

This theorem does not assume bounded diameter, $\pi_2 = 0$, a class-square cost, or base injectivity radius. It allows genuine rank-one or rank-two collapse; Section 6 gives an exact rank-two example. The classical rank/model test (2) is not proved to hold everywhere. The exceptional-volume estimate is relative to the whole manifold, not to each arbitrarily small thin component.

2. The covering-average lemma: multiplicity is not free

Lemma 1. Suppose $\mathrm{Ric} \geqslant -3g$, $\pi : U \to B_R(x)$ is a connected normal Riemannian covering, and $4b < R$, $b \leqslant 1$. For any nonnegative measurable function F on $B_R(x)$,

$$\fint_{B_b(\tilde{x})} \pi^* F \leqslant C_1 \fint_{B_b(x)} F, \qquad C_1 \leqslant \frac{V_{-1}(3b)}{V_{-1}(b)} \leqslant C(4). \tag{5}$$

The constant does not depend on the number of sheets, a base-ball volume, or an injectivity radius. Both sides are **averages**.

Proof. Choose a Dirichlet fundamental domain based at $\tilde{x}$. Its part D_b projecting to $B_b(x)$ has volume $\mathrm{Vol}\, B_b(x)$ and lies in $B_b(\tilde{x})$, apart from measure-zero choices on its boundary. Indeed a minimizing base geodesic from x to any point of $B_b(x)$ lifts inside that ball, and the Dirichlet representative is no farther.

If a point of $B_b(x)$ has m lifts in $B_b(\tilde{x})$, each is a translate of its representative in D_b. Its deck transformation moves $\tilde{x}$ by less than $2b$. Denote the number of such deck transformations by N. The sets γD_b for these transformations are disjoint up to measure zero and contained in $B_{3b}(\tilde{x})$. Therefore

$$m \leqslant N \leqslant \frac{\mathrm{Vol}\, B_{3b}(\tilde{x})}{\mathrm{Vol}\, B_b(x)}. \tag{6}$$

The balls have compact closure away from the boundary of the cover; proper discontinuity makes the relevant orbit count finite. The area formula and (6) bound the lifted integral by $\mathrm{Vol}\, B_{3b}(\tilde{x}) \fint_{B_b(x)} F$. Divide by $\mathrm{Vol}\, B_b(\tilde{x})$ and apply the local Bishop–Gromov comparison, which is valid since $3b < R$. This proves (5). □

In contrast, an estimate for an unnormalized integral lifts with the multiplicity in (6), which can diverge. A global small average does not say that every local base average is small. The next step quantifies exactly this second issue rather than suppressing it.

3. Global gauge error gives small local averages off a small set

The inherited actual-form theorem gives global forms $\omega_{\pm} \in \Omega^{\pm}(M)$, with $\|\omega_{\pm}\|_\infty \leqslant 2\sqrt{3}$. Put

$$F = \sum_{\sigma=\pm} \Big[(|\omega_\sigma|^2 - 2)^2 + |\nabla \omega_\sigma|^2 \Big]. \tag{7}$$

Use the smoothing tolerance $K/4$ for each form and write $\kappa_\sigma = r + q_\sigma$, so $\kappa_+ + \kappa_- = K$. The actual estimates give

$$\fint F \leqslant 734K + 8\sqrt{K} \leqslant 750\sqrt{K}. \tag{8}$$

The first-order bound and the uniform amplitude, not the second-order Hodge estimate, are used. There is no hidden negative-square cost.

For any fixed $b \leqslant 1$, let $\mathscr{A}_b F(x) = \fint_{B_b(x)} F$. Local volume doubling under the Ricci lower bound implies

$$\int_M \mathscr{A}_b F(x)\, d\mu(x) \leqslant C_2 \int_M F. \tag{9}$$

To see this explicitly, if $d(x,y) < b$, then $B_b(y) \subset B_{2b}(x)$, hence $\operatorname{Vol} B_b(y) \leqslant C_2 \operatorname{Vol} B_b(x)$. Fubini applied to the ball-averaging kernel now proves (9). Consequently, for every $t > 0$,

$$\frac{\operatorname{Vol}\{x : \mathscr{A}_b F(x) > t\}}{V} \leqslant \frac{750 C_2 \sqrt{K}}{t}. \tag{10}$$

Take $t = K^{1/4}$. This proves (3). At every remaining point, Lemma 1 transfers the small base average to its local universal cover:

$$\fint_{B_b(\tilde{x})} \pi_x^* F \leqslant C_1 K^{1/4}. \tag{11}$$

The potentially enormous covering multiplicity has cancelled against the correct local volume normalization. No inequality asserting $\int \pi_x^* F \leqslant C \int_M F$ is used.

4. Maximal rank supplies regularity; the forms identify the limit

For $x \in \mathscr{G}_\rho$, apply Naber–Zhang, Proposition 6.4, to $\rho^{-2} g$. Since $\delta \leqslant 1/2$, the Ricci tensor norm of g is at most three, and rescaling preserves the required bound. The conclusion supplies a uniform harmonic radius at the lift of x in the universal cover of $B_\rho(x)$, at least $h_N \rho$. This is a classical theorem used with its rank condition, not a new claim that Ricci bounds alone give regular local covers.

Choose b to be a fixed sufficiently small multiple of $h_N \rho$, with $4b < \rho$. The covering metric has controlled harmonic coordinates on a neighbourhood larger than $B_b(\tilde{x})$. The harmonic metric equation and bounded Ricci give uniform finite $W^{2,p}$ bounds on smaller charts and hence $C^{1,\beta}$ precompactness for every $\beta < 1$. No derivative of Ricci is needed.

We give the argument for the qualitative uniform modulus in Theorem A. If it failed, there would be good points with $\delta_j \to 0$, $K_j^{1/4} \to 0$, and controlled cover charts that stay a fixed distance from product geometry. The charts have a convergent subsequence. In harmonic coordinates

$$g^{ab} \partial_a \partial_b g_{ij} = -2 \operatorname{Ric}_{ij} + Q_{ij}(g, \partial g). \tag{12}$$

Pointwise $\operatorname{Ric} + g \to 0$ makes the limit a smooth exact negative Einstein metric. Subtracting its equation improves convergence to strong finite $W^{2,p}$, and therefore to the prescribed $C^{1,\alpha}$ on still smaller charts.

Lift the actual forms to these charts. Their uniform bounds and (11) give fixed-chart H^1 bounds. Rellich compactness and convergence of connections yield two smooth parallel limiting forms, one self-dual and one anti-self-dual, both of norm $\sqrt{2}$. The associated parallel complex structures commute. Their product is a symmetric trace-zero involution, with two rank-two parallel eigenspaces. Local de Rham splitting and $\operatorname{Ric} = -g$ make both factors hyperbolic surfaces of curvature -1. On a fixed smaller ball this is exactly (4), contradicting the assumed failure of uniform closeness. This proves Theorem A. □

The limit assertion uses neither global compactness nor an ALE theorem. It also does not identify the limiting local deck transformations across different points or across the exceptional set. That missing global information is precisely why Theorem A is not a global realization result.

5. Why this changes the scope, and where it stops

The earlier bounded-diameter bilateral theorem obtains one compact smooth limit and hence an exact metric on the original manifold. The present theorem removes diameter and **base noncollapse** at the cost of a different conclusion: quantitative product geometry in local covers on the maximal-rank bulk. The two conclusions are not interchangeable.

There are two distinct failure mechanisms left visible:

1. A point can fail the rank/model condition (2). Small global gauge deficit does not prove maximal nilpotent rank or a smooth collapsed base. Naber–Zhang's singular-fibre examples explain why that topology cannot simply be omitted from their theorem.
2. A point can belong to $\mathscr{E}$. The global bounds control its volume, not its topology, its monodromy, or the tensor modes needed to solve the global Einstein equation.

Even if (2) holds everywhere and the exceptional-volume bound tends to zero, no assertion here controls a full global tensor inverse or glues local exact product limits into a single metric on M. The diameter-free repair criterion cannot be invoked by replacing its base-ball Moser estimate with an unexplained cover estimate: the inverse and the global residual norm need their own covering-compatible control. Theorem A supplies genuine collapsed local geometry, not that missing global solver.

6. Actual collapse tests and a sharp failure of unnormalized transfer

6.1 Exact bilateral saturation allows rank-two thin geometry

Take two closed hyperbolic surfaces of fixed genera at least two, each with a separating geodesic of length $\ell \to 0$, and their product metric g_ℓ. The canonical classes of the two product complex structures are actual in both orientations, and

$$\delta_\ell = q_{+,\ell} = q_{-,\ell} = 0. \tag{13}$$

Near the two short geodesics the metric is exactly

$$dt_1^2 + \ell^2 \cosh^2(t_1) d\theta_1^2 + dt_2^2 + \ell^2 \cosh^2(t_2) d\theta_2^2, \tag{14}$$

where each θ_i has period one. Collar widths tend to infinity. At their common waist, fixed-radius base-ball volumes are $O(\ell^2)$. The balls converge to Euclidean two-dimensional balls, and the fibre group in (2) is $\mathbb{Z}^2$, of maximal rank $4-2$. Unwrapping the two circles gives exact hyperbolic-product geometry. These are real negative Einstein and gauge-saturated examples excluded by any uniform base-ball volume lower bound, but covered by Theorem A.

This is not a new Einstein construction. It is an exact test showing that the theorem's changed input regime includes genuine collapse, not merely a reformulated noncollapsed condition.

6.2 Absolute integral errors need not stay small after rewinding

On the same actual metrics choose a fixed nonzero nonnegative smooth function $f(t_1, t_2)$, supported in $|t_i| < b/10$, independent of θ_1, θ_2, and extend it by zero. Then

$$\int_M f \, d\mu_{g_\ell} = \ell^2 \int f \cosh t_1 \cosh t_2 \, dt_1 dt_2 \to 0. \tag{15}$$

On the local universal cover use $u_i = \ell\theta_i$. The lifted metric is $dt_1^2 + \cosh^2(t_1)du_1^2 + dt_2^2 + \cosh^2(t_2)du_2^2$, independent of ℓ. Its lifted function is the same nonzero $f(t_1, t_2)$. Its integral over a fixed suitably chosen lifted ball is bounded below by a positive constant. Thus no universal estimate of the form

$$\int_{B_b(\tilde{x})} \pi^* f \leqslant C \int_M f \tag{16}$$

can hold, even when the underlying metrics have exact Einstein and bilateral gauge equality. This counterexample concerns an inference between integral norms. The function f is not asserted to be an actual monopole defect on the equality metrics, where those defects are zero. It pinpoints why a proof based only on their global integral bounds cannot freely lift such bounds without a local-normalization step.

6.3 Local and global universal covers are different objects

For example, a Hopf circle in S^3 is contractible in S^3, but represents the infinite cyclic fibre group over a small disk in the Hopf base. A local cover of that tube unwraps the circle; the global universal cover does not. Taking a product with S^1 gives a four-dimensional example with $\pi_2 = 0$ exhibiting this distinction. This elementary example is not claimed to satisfy the negative Einstein/gauge hypotheses. It shows why even $\pi_2(M) = 0$ does not justify embedding every rewound local ball into $\widetilde{M}$. Theorem A works entirely on the local cover and never makes that inference.

7. Source and status

- Naber–Zhang, *Topology and epsilon-regularity Theorems on Collapsed Manifolds with Ricci Curvature Bounds*, Proposition 6.4, supplies the maximal-rank harmonic-radius theorem on the local universal cover; its Proposition 6.5 requires the exact Einstein equation or Ricci-derivative control for curvature bounds. Only Proposition 6.4 is used here. We specialize its model to Euclidean space so its constants do not hide a varying limit-space dependence.
- The actual-form theorem supplies (7)–(8); bilateral realization gives the opposite-handed form algebra. The present theorem does not require the second-order form estimate.
- The preserved collar construction, Section 2, records the classical hyperbolic collar formula and its primary reference. The product tests above use these classical metrics, not a new gluing construction.

The new project bridge is the combination of actual gauge supply, covering averages, and maximal-rank collapsed regularity. Its bad-set bound is proved by a direct averaging argument; its local geometric modulus is qualitative compactness. It neither proves maximal rank from saturation nor settles global Einstein existence in the collapsing regime.